For Solar Power:

$$P_{PV}(t) = P_{rated}\, D_{PV}\left(\frac{S(t)}{S_{ref}}\right)\left[1 + \alpha_p\left(T_{cell}(t) - T_{ref}\right)\right] \quad (1)$$

The cell temperature of PV module was found from Equation (2), where $T_a(t)$ is the ambient temperature. $T_{cell,NOCT}$ *and* $T_{a,NOCT}$ is the Nominal Operating Cell Temperature (NOCT) and NOCT at the ambient temperature, respectively. In addition, S_{NOCT} is the incident solar irradiation at NOCT (800 W m^{-2}), η_{PV} is the panel efficiency (20.1%), $\tau\alpha$ is the solar transmittance-absorptance of the PV module, which is assumed to be 0.9 in HOMER platform [45].

$$T_{cell}(t) = T_a(t) + S(t)\left(\frac{T_{cell,NOCT} - T_{a,NOCT}}{S_{NOCT}}\right)\left(1 - \frac{\eta_{PV}}{\tau\alpha}\right) \quad (2)$$

PV output power was determined from Equation (1), where P_{rated} is the PV rated power (327 W at STC), D_{PV} is the derating factor of the module (88%), $S(t)$ is the incident solar irradiation on the PV, S_{ref} is the incident solar irradiation at STC (1000 Wm^{-2}), α_P is the temperature coefficient of power ($-0.38\%.K^{-1}$), $T_{cell}(t)$ is the PV cell temperature, and T_{ref} is the reference cell temperature at STC (25 °C) [44].

Nominal power, PPV 327 W
Panel efficiency, hPV 21.4%
Rated voltage, Vmpp 54.7 V
Rated current, Impp 5.98 A
Open circuit voltage, Ioc 64.9 V
Short circuit current, Isc 6.46 A
Power temperature coefficient, aP 0.35%. C -1
NOCT 45 C ± 2 C

Components Type	Description	Capital cost ($)	Replacement cost ($)	O&M cost ($.kW^{-1}.yr^{-1})
PV module [56]	327 W	1300 kW^{-1}	0	10

For Wind Power:

Rated speed, Vr = 8.5 ms-1
Cut in speed, Vcut-in = 2.7ms-1
Cut-out speed, Vcut_out = 20 ms-1
Hub height, H = 36 m

Since velocity is highly variable due to the climatic conditions, $P_{WT}(t)$ is a function of wind speed and is estimated by Equation (4) [47].

$$P_{WT}(t) = \begin{cases} 0, & V(t) \le V_{cut-in} \text{ or } V(t) \ge V_{cut-out} \\ P_r\left(\dfrac{V^3(t) - V^3_{cut-in}}{V^3_r - V^3_{cut-in}}\right), & V_{cut-in} < V(t) < V_r \\ P_r, & V_r < V(t) \le V_{cut-out} \end{cases} \quad (4)$$

where, $V(t)$, V_r, V_{cut-in}, $V_{cut-out}$, and P_r refer wind velocity (ms^{-1}), rated wind velocity (ms^{-1}), cut-in wind velocity (ms^{-1}), cut-out wind velocity (ms^{-1}), rated power output (kW), respectively. The wind velocity at turbine hub altitude is obtained from Equation (5) [47], whereby, H is the wind turbine hub altitude (m), and V_{ref} is the

reference velocity (ms^{-1}) at a reference height H_{ref} (m) and α is the Hellman coefficient. The value of this coefficient varies from 0.10 to 0.25 [48].

$$\frac{V_H}{V_{ref}} = \left(\frac{H}{H_{ref}}\right)^{\alpha} \quad (5)$$

Wind turbine [13,58]	20 kW	2500 kW^{-1}	1750 kW^{-1}	50	20

Or,

Wind turbine [34,77,78]	Nominal capacity, P_r	25 kW
	Cut-in wind speed, V_{ci}	2.75 m/s
	Rated wind speed, V_r	9 m/s
	Cut-out wind speed, V_{co}	20 m/s
	Hub height, h	36 m
	Capital cost	2300 $/kW
	O&M cost in the first year	3% of capital cost
	Lifetime	20 years

For Inverter:

Inverter [55] Rated power, Pinv 1 kW
Conversion efficiency, hinv 95%

Inverter [56]	1 kW	300	300	0	15

Or,

Converter	Rated power	115	kW
	Rectifier(η_{rec}) and inverter(η_{inv}) efficiency	90	%
	Replacement cost (per kW)	127	$/kW
	O & M cost (per kW)	1	$/yr
	Life time	20	years

For Diesel Generator:

2.2.2. Diesel generator modelling

In this study, a diesel generator of 50 kW capacity is selected. HOMER assumes that the fuel consumption curve for a diesel generator is a straight line. The fuel consumption for a 50 kW generator is calculated using Equation (3) [50], whereby a is the co-efficient due to the fuel curve intercept (1.65 l/h/kW) and b is the slope of the fuel curve (0.267 l/h/kW) [9]. In addition, R_{DG} is the diesel generator's rated capacity (50 kW), and $P_{DG}(t)$ is the power generation at any time interval.

$$F_{DG}(t) = aR_{DG} + bP_{DG}(t) \tag{3}$$

The costs associated with capital investment, replacement, and O&M are 370$/kW, 296$/kW, and 0.05$/h/kW of operation, respectively [56], where the lifetime of the generator is chosen as 15000 h [57]. The fuel cost (diesel fuel) is considered to be 0.70 $/l.

or,

$$C_F = P_f F_{gen} \tag{2}$$

In the last equation, P_f is the fuel price ($/L), and F_{gen} is the fuel consumption rate which depends on the output power of the generator. It can be expressed by [38]

$$F_{gen} = f_{gen_0} p_g + f_{gen_1} p_r \tag{3}$$

where p_g and p_r are the generator output and rated power respectively. Also, f_{gen_0} is the intercept coefficient of the fuel curve which equals 0.246 L/kWh, and f_{gen_1} is the fuel curve slope which equals 0.08145 L/kWh.

Component	Capital ($/kW)	O&M ($/kW/yr.)	Replacement ($/kW)	Fuel ($/L)
DG (Atallah et al. 2020)	500	0.03 $/h	500	0.77